

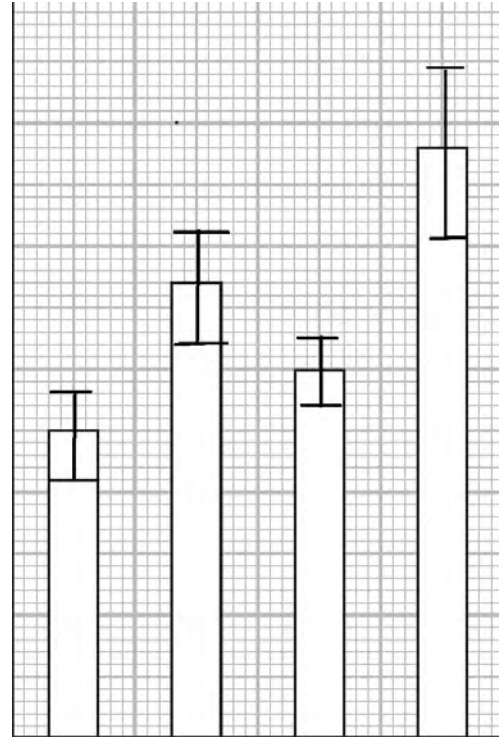
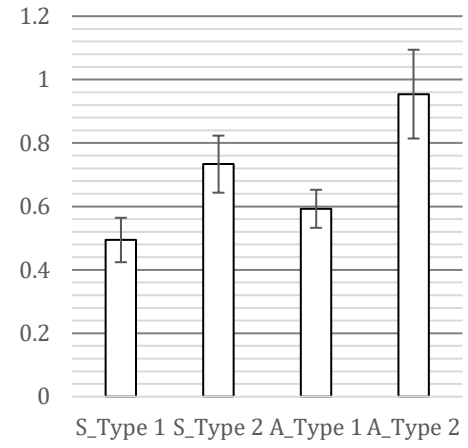
Question Number	Answer	Additional Guidance	Mark
1(a)	An answer that includes the following points: - because the antioxidants will {reduce (the action of) / stabilise} free radicals / reduce oxidative stress (1) - therefore damage to {(endothelial) cells / endothelium} will be reduced (1) - so there will be less chance of plaque build-up / formation of atheroma (1)	accept attack, remove free radicals accept (endothelial) lining of BV not atherosclerosis	(3)

Question Number	Answer	Additional Guidance	Mark
1(b)	A description that includes the following points: 1. use equal {size pieces / masses} of spinach (for both plants) (1) 2. standard extraction method described (1) 3. measure out {equal / same / stated} volume of (standard) DCPIP solution (1) 4. add spinach (extract) until no blue colour remains / becomes colourless / decolourises (1) 5. record volume of extract used (1) 6. use of calibration curve / standard solutions (1)	e.g. grinding time / volume of water measure out {equal / same / stated} volume of spinach extract add DCPIP solution until becomes permanently blue record volume of DCPIP used e.g mass of vitamin C in fruit juice sample = mass of vitamin C to decolourise 1cm³ of DCPIP × volume of sample required to decolourise 1cm³ of DCPIP	(5)

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	A calculation with the following steps: • correct calculation of numerator and denominator (1) • correct division of numerator by denominator and square root found (1) • answer correctly rounded to two decimal places (1)	Allow ecf 0.2 and 11 (accept 12-1) 0.0181 and 0.13483 0.14, allow 0.13	(3)

Question Number	Answer	Additional Guidance	Mark
1(c)(ii)	A graph showing the following features: • y axis with fully labelled linear scale (1) • x and y axes correctly labelled with names of variables in each case, and units on y. X could be labeled using a key (1) • SDs plotted correctly (1)	Scale values should be indicated at equal increments, minimum 1 Allow $\frac{1}{2}$ square tolerance	

Mean
vitamin C
content
mg g⁻¹



Question Number	Answer	Additional Guidance	Mark
1(c)(iii)	An answer including the following points: • a correct comparison of means for both {soil types / seasons} / all means different (1) • difference between soil types is significant as SDs do not overlap (for both seasons) (1) • difference between seasons for same soil type are not significant as SDs overlap (1)	e.g. autumn higher than spring (for both types) / type 2 higher than type 1 (in both seasons) ecf if SD for Autumn type 2 is plotted too big ecf if SD for Autumn type 2 is plotted as too small if they say autumn diff is significant due to overlap	(3)